Remittance Frictions, Migrant Heterogeneity, and Transfer Behavior

Research Proposal

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April 2026

Contents

1 Motivation and Research Question

Remittances are macroeconomically large, distributionally central, and persistently expensive. Recorded flows to low- and middle-income countries exceed $650 billion annually, and average transfer costs remain well above the SDG 10.c target of 3 percent (World Bank, 2023; World Bank RPW, 2023). Evidence from the BIS and the World Bank shows that these flows are resilient but shock-sensitive, with meaningful corridor heterogeneity during and after the pandemic (BIS, 2021; World Bank, 2023). At the household and corridor level, transaction costs, exchange-rate margins, and channel access constraints shape formal remittance behavior (Ahmed et al., 2021; Freund and Spatafora, 2008).

The central question is:

> *How do fixed and variable remittance frictions and macro-level shocks shape whether migrants remit and how much they send, and can a single heterogeneous-agent framework link those two margins to remittance patterns across regions and over time?*

The paper adapts the logic of heterogeneous-firm trade models to remittances: migrants differ in income and altruism/obligation intensity, remitting requires paying a fixed access cost and a variable ad-valorem cost, only migrants with sufficiently high net benefit remit, and among those who do, they select which recipient types to serve. The main contribution is a unified framework for policy counterfactuals on cost reduction, digitalization, and transfer-market regulation, with the extensive-intensive decomposition disciplined by micro data on both margins.

2 Literature Review and Contribution

This paper contributes to five literatures, drawing on both labour economics and trade theory.

Remittance microeconomics: Classic work establishes altruism, exchange, and insurance motives (Lucas and Stark, 1985; Stark and Bloom, 1985; Yang and Choi, 2007), but typically does not map coefficients to structural objects usable for counterfactuals or aggregate predictions. Most of the literature agrees on the safety-net role of remittances, but the relative importance of different motives remains an open question with mixed evidence. Bollard et al. (2011) provide cross-country micro evidence on remittance determinants and show substantial heterogeneity in remitting behavior across migrant groups, but do not embed that heterogeneity in a structural framework suitable for cost counterfactuals. Rapoport and Docquier (2006) survey the theoretical and empirical state of the field and identify the gap between micro motives and aggregate predictions.

Remittance costs and formalization: Higher transfer costs reduce formal remittance flows and can shift activity to informal channels (Freund and Spatafora, 2008;

Ahmed et al., 2021). This matters because formal cost wedges remain economically large relative to transfer size (Ratha, 2006; World Bank RPW, 2023). Direct evidence on remittance-cost elasticities is still relatively limited. Gibson et al. (2006) provide one of the clearest benchmarks, showing in the New Zealand–Tonga corridor that remittance flows respond negatively to transfer costs. Taken together, the existing evidence suggests economically meaningful cost sensitivity, but it does not pin down a single elasticity benchmark that travels cleanly across settings.

Remittance responses to exchange-rate shocks: Yang (2008) provides the key identification benchmark in this literature: exploiting exchange-rate shocks induced by the Asian financial crisis, he shows that favorable exchange-rate movements for Filipino migrants increase remittance receipts and related household investment. Yang and Choi (2007) extend this to test the insurance motive. This paper follows the Yang (2008) logic but adapts it to the current region-year design: exchange-rate variation enters through a region-level effective index (fixed-weight aggregation of country-level exchange rates within each birth region), not through fully disaggregated corridor-year estimation.

Migration and spatial equilibrium: Albert and Monras (2022) show that immigrants devote part of their income to expenditures in the country of origin, which provides a useful conceptual benchmark for treating origin-household resources as part of the migrant's allocation problem.

Trade-style heterogeneous-agent structure: The empirical architecture follows the logic of selection and intensive-margin decomposition in Melitz (2003); Helpman et al. (2008); Chaney (2008), but the economic object is intrafamily transfer behavior rather than firm export sales. The key mapping is that migrants with heterogeneous altruism/obligation intensity select into remitting (analogous to productivity-based export selection), and conditional on remitting, choose transfer amounts to multiple recipient types (analogous to firms choosing which destinations to export to and at what volume).

Gap addressed: The paper delivers a single framework linking participation, recipient-type selection, amount decisions, and aggregate remittance responses to transfer frictions, with explicit discipline on what is identified in the current data environment. Relative to the experimental literature, the contribution is a unified structural framework that can rationalize the extensive-intensive decomposition and generate counterfactuals for cost-reduction policies. Relative to the structural trade literature, the contribution is adapting the multi-destination selection logic to a new domain where micro data on both margins are directly observed.

3 Conceptual Framework

3.1 Preferences

Consider migrant i from origin region r in year t. The migrant faces a utility maximization problem under a CES aggregator over local consumption (where the migrant resides) and the total resources available to the origin household (where remittances are sent):

$$U_{irt} = \left[\alpha(c^L_{irt})^{\frac{\sigma-1}{\sigma}} + (1-\alpha)\left(\phi_{ir}(R_{irt} + y^O_{rt})\right)^{\frac{\sigma-1}{\sigma}}\right]^{\frac{\sigma}{\sigma-1}}, \tag{1}$$

where $\phi_{ir} > 0$ denotes the altruism/obligation intensity parameter, which may reflect family ties, cultural norms, number of dependents, or network-mediated social expectations. This parameter is analogous to the productivity parameter in Melitz (2003): it is the source of micro-level heterogeneity that drives selection into remitting. c^L_{irt} is the migrant's consumption in the destination country, R_{irt} is the amount sent in remittances, and y^O_{rt} is the origin household's own income (observable to the migrant). Origin-household total resources are $R_{irt} + y^O_{rt}$, which encompasses both consumption and savings at origin, since we cannot observe how the origin household allocates received funds, we treat this as a single composite good. The presence of y^O_{rt} creates a role for origin-country economic conditions: negative income shocks at origin (lower y^O_{rt}) increase the marginal utility of remittances, generating an insurance motive consistent with Yang and Choi (2007). We cannot separately identify whether the motive for remitting is pure altruism, exchange, insurance, or risk-sharing self-interest as in Lucas and Stark (1985), the parameter ϕ_{ir} captures the net obligation intensity from all sources. σ is the elasticity of substitution between local consumption and origin-household resources.

The budget constraint that the individual faces is:

$$c^L_{irt} + f_{rt}\mathbf{1}\{R_{irt} > 0\} + (1+\tau_{rt})R_{irt} \leq w_{irt}, \tag{2}$$

where f_{rt} is a fixed remitting cost, capturing one-time or per-transaction costs of accessing a remittance channel (analogous to the fixed export cost in Melitz (2003)), τ_{rt} is the ad-valorem cost (analogous to the iceberg cost structure in Melitz (2003)), w_{irt} is the wage, and R_{irt} is the amount sent in remittances.

3.2 Optimal Remittance Decision

Conditional on remitting $R_{irt} > 0$, the migrant maximizes utility subject to the budget constraint. Let $\rho = (\sigma - 1)/\sigma$ and write the Lagrangian:

$$\mathcal{L} = \alpha(c^L)^\rho + (1-\alpha)\phi(R + y^O)^\rho + \lambda\left((w - f) - c^L - (1+\tau)R\right).$$

First-order conditions imply

$$\alpha\rho(c^L)^{-1/\sigma} = \lambda, \qquad (1-\alpha)\phi\rho(R+y^O)^{-1/\sigma} = \lambda(1+\tau),$$

so the optimal ratio of origin-household resources to local consumption is

$$\frac{R_{irt}+y^O_{rt}}{c^L_{irt}} = \left[\frac{(1-\alpha)\phi_{ir}}{\alpha(1+\tau_{rt})}\right]^\sigma \equiv A_{irt}. \tag{3}$$

Using the budget constraint $c^L + (1+\tau)R = w - f$:

$$R^*_{irt} = \frac{(w_{irt} - f_{rt} + (1+\tau_{rt})y^O_{rt})A_{irt} - y^O_{rt}}{1+(1+\tau_{rt})A_{irt}}, \qquad c^{L*}_{irt} = \frac{w_{irt} - f_{rt} + (1+\tau_{rt})y^O_{rt}}{1+(1+\tau_{rt})A_{irt}}. \tag{4}$$

When origin-household income y^O_{rt} is small relative to the migrant's net resources ($w_{irt} - f_{rt}$), the expressions simplify to the standard CES form $R^* \approx (w-f)A/(1+(1+\tau)A)$.

The effective expenditure share devoted to remitting is:

$$\psi^x_{irt} \equiv \frac{(1+\tau_{rt})R^*_{irt}}{w_{irt}-f_{rt}} \approx \frac{(1+\tau_{rt})A_{irt}}{1+(1+\tau_{rt})A_{irt}} \tag{5}$$

where the approximation holds when $y^O_{rt}/(w_{irt}-f_{rt}) \to 0$. Under this approximation:

$$\tilde{\psi}_{irt} \equiv \frac{A_{irt}}{1+A_{irt}} \quad \Rightarrow \quad \log\left(\frac{\tilde{\psi}_{irt}}{1-\tilde{\psi}_{irt}}\right) = \sigma\log\phi_{ir} + \sigma\log\left(\frac{1-\alpha}{\alpha}\right) - \sigma\log(1+\tau_{rt}), \tag{6}$$

which gives the linear-in-logs intensive-margin object that maps directly to the empirical specification. In practice, origin-household income variation is absorbed by region-year fixed effects ($\mu_r + \lambda_t$) in the estimation equations, since y^O_{rt} varies at the region-year level.

Participation condition. The migrant remits iff $R^*_{irt} > 0$, which requires

$$(w_{irt} - f_{rt} + (1+\tau_{rt})y^O_{rt})A_{irt} > y^O_{rt},$$

or equivalently, the migrant's optimal allocation to origin exceeds what the origin household already has. When $y^O_{rt} = 0$ (no origin income), this reduces to the standard $A_{irt} > 0$ condition (always satisfied for $\phi_{ir} > 0$), and the binding constraint is the fixed cost f_{rt}. More generally, the participation decision defines a Melitz-style cutoff:

$$\Delta U_{irt}(\phi, w, \tau, f, y^O) \equiv U^R_{irt} - U^0_{irt} \gtrless 0,$$

so remitting occurs iff $\Delta U_{irt} \geq 0$, with cutoff $\phi^*(w,\tau,f,y^O)$ and comparative statics: $\partial\phi^*/\partial\tau > 0$, $\partial\phi^*/\partial f > 0$, $\partial\phi^*/\partial w < 0$, and $\partial\phi^*/\partial y^O > 0$. The last comparative static captures the insurance motive: when origin income y^O_{rt} is high, only high-altruism migrants remit, while negative origin shocks lower the cutoff and bring more migrants into the remitting population.

This yields three testable margins:

- **Extensive margin:** remitting participation cutoff in (ϕ, w, y^O) space.
- **Intensive margin:** conditional transfer share and amount response to τ_{rt} and f_{rt}.
- **Recipient-type selection margin:** described in Section 3.3 below.

3.3 Extension: Multiple Recipient Types as Destinations

The baseline model treats remittances as a single aggregate transfer. However, SIPP reports transfers to four distinct recipient types: parents (`TPARTOTAMT`), other relatives (`TORTOTAMT`), non-relatives (`TNRTOTAMT`), and children outside the household (`TKIDTOTAMT`). This disaggregation enables a tighter mapping to the Melitz (2003) multi-destination structure: each recipient type $j \in \{1, \ldots, J\}$ is a "destination" to which the migrant may or may not send transfers, facing a recipient-specific altruism weight ϕ_{ij} and a type-specific fixed cost f_{jrt}.

Formally, extend the utility function to

$$U_{irt} = \left[\alpha(c^L_{irt})^\rho + \sum_{j=1}^{J}(1-\alpha_j)\left(\phi_{ij}(R_{ijrt} + y^O_{jrt})\right)^\rho\right]^{1/\rho}, \tag{7}$$

where α_j governs the weight on recipient type j, ϕ_{ij} is the migrant-recipient altruism weight, R_{ijrt} is the transfer to type j, and y^O_{jrt} is the origin income of recipient group j. The budget constraint becomes

$$c^L_{irt} + \sum_{j=1}^{J}\left[f_{jrt}\mathbf{1}\{R_{ijrt} > 0\} + (1+\tau_{rt})R_{ijrt}\right] \leq w_{irt}.$$

Conditional on the set of active recipient types $\mathcal{J}_i \subseteq \{1, \ldots, J\}$, the within-type allocation follows the same CES logic as the baseline, with type-specific $A_{ijrt} = [(1-\alpha_j)\phi_{ij}/(\alpha(1+\tau_{rt}))]^\sigma$. Selection into type j requires the utility gain from sending to j to exceed the fixed cost f_{jrt}, defining a type-specific cutoff $\phi^*_j(w, \tau, f_j, y^O_j)$.

This yields two additional testable predictions:

1. **Recipient-type extensive margin:** the number of active recipient types should decrease with remittance costs τ_{rt} and f_{rt} (region-level in current implementation), and increase with migrant income w_{irt}. Higher-altruism migrants send to more recipient types, analogous to higher-productivity firms exporting to more destinations.
2. **Pecking order:** recipient types should be activated in a predictable order determined by the altruism ranking ϕ_{ij}. If parents have the highest average ϕ, they

should be served first, with other relatives, children, and non-relatives added as the altruism/income threshold is exceeded.

The baseline empirical strategy aggregates across recipient types (using the strict proxy as the primary specification). The multi-recipient extension is tested through recipient-type-specific participation equations and by examining the correlation between the number of active transfer types and migrant characteristics.

4 Data, Measurement, and Identification Discipline

4.1 Micro Data and Unit of Analysis

The micro data used for the empirical analysis come from the 2019–2024 Survey of Income and Program Participation (SIPP). Current implementation is **person-year** at the micro stage and **birth-region by year** at the aggregate stage.

In practice, the design combines a nativity split (US-born versus foreign-born) with six foreign-born birth-region groups (Europe, Asia-Pacific, Americas-Caribbean, Africa, Oceania, and Other). The current birth-region mapping is therefore a coarse proxy for corridor origin, and current cost data are region-year averages built from the main corridors represented in each region. As a result, the present specification is not yet structurally identified at the bilateral corridor level. Inference is additionally constrained by a very small effective number of cost clusters in current runs, so standard errors and hypothesis tests are treated as indicative diagnostics rather than decisive evidence.

4.2 Remittance Proxy: Definitions and Validity

SIPP does not directly tag international remittances, but it reports annual transfer amounts by recipient type (`TPARTOTAMT`, `TORTOTAMT`, `TNRTOTAMT`, `TKIDTOTAMT`), which can be used as remittance proxies.

- `TPARTOTAMT`: transfers to parents outside the household.
- `TORTOTAMT`: transfers to other relatives outside the household.
- `TNRTOTAMT`: transfers to non-relatives.
- `TKIDTOTAMT`: transfers to children outside the household.

The paper uses two proxy layers:

- **Strict proxy (baseline):** `TPARTOTAMT + TORTOTAMT`.
- **Broad proxy (robustness):** strict proxy + `TNRTOTAMT + TKIDTOTAMT`.

The proxy choice is a key design decision that trades off remittance relevance (strict proxy) against coverage and potential noise (broad proxy). The objective is to capture remittance-relevant behavior rather than perfectly isolate international remittances from all domestic transfers. Some contamination is acceptable if it is not systematically correlated with transfer-cost variables, and this is assessed through nativity and regional validation exercises.

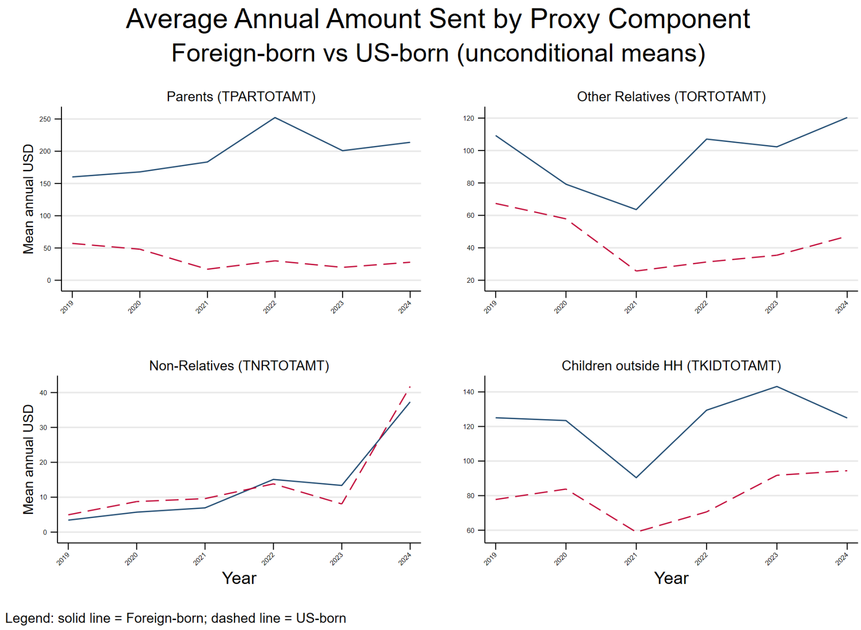


Figure 1: Transfer Components by Nativity (Foreign-born vs US-born)

Figure 1 shows that the largest nativity gaps are in transfers to parents and other relatives, which are precisely the components used in the strict proxy. This pattern supports the baseline proxy choice while still leaving domestic-transfer contamination as an explicit limitation.

Shares are defined with **annual earnings** (sum of monthly `TPEARN`) in the denominator, not total income including transfer components. For evolution plots, the estimand is ratio-of-totals:

$$\text{Share}_t = \frac{\sum_i R_{it}}{\sum_i w_{it}}, \tag{8}$$

because mean-of-ratios is dominated by low-denominator outliers.

Bounded-sample rule for conditional-share plots:

- annual earnings $\geq \$5{,}000$,
- at least 6 monthly earnings observations,
- share in $(0, 1]$.

I also evaluate differences in sending rates between foreign-born and US-born individuals:

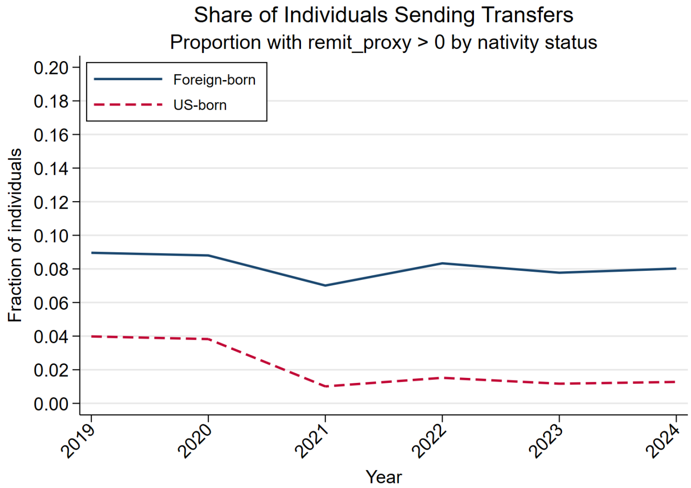


Figure 2: Sending Rates by Nativity

In the pooled sample, sending rates are substantially higher for the foreign-born group than for the US-born group, which is consistent with the proxy capturing meaningful international-remittance behavior.

Regional heterogeneity is also informative:

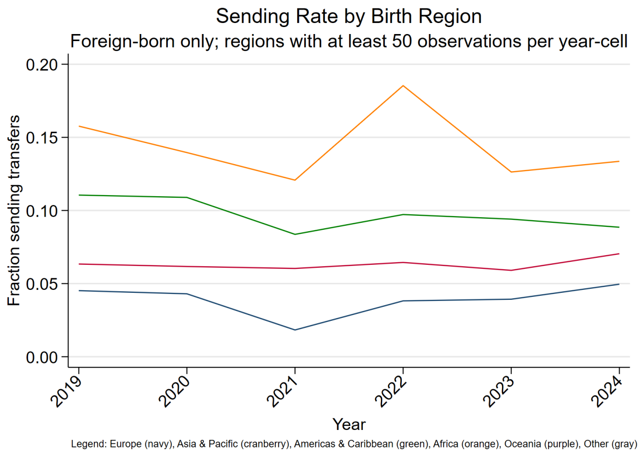


Figure 3: Sending Rates by Foreign-Born Region of Birth

Figure 3 shows higher participation in regions more exposed to remittance-intensive origin countries, which is again consistent with the proxy design. At the same time, these are validation patterns rather than definitive proof of perfect measurement.

Finally, we examine transfer-share dynamics:

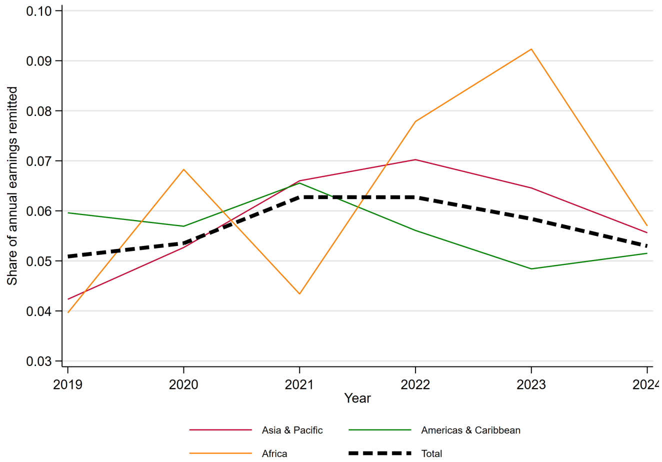


Figure 4: Evolution of Transfer Share of Earnings (Strict Proxy)

Under the strict proxy, aggregate conditional shares are materially below the 15% benchmark that the UN remarks and remain in the mid-single digits. The broad proxy produces higher shares, but the strict-versus-broad spread is interpreted as a measurement sensitivity band, not as a pure remittance-versus-non-remittance gap, given that the broad proxy includes some components that are likely to contain domestic transfers:

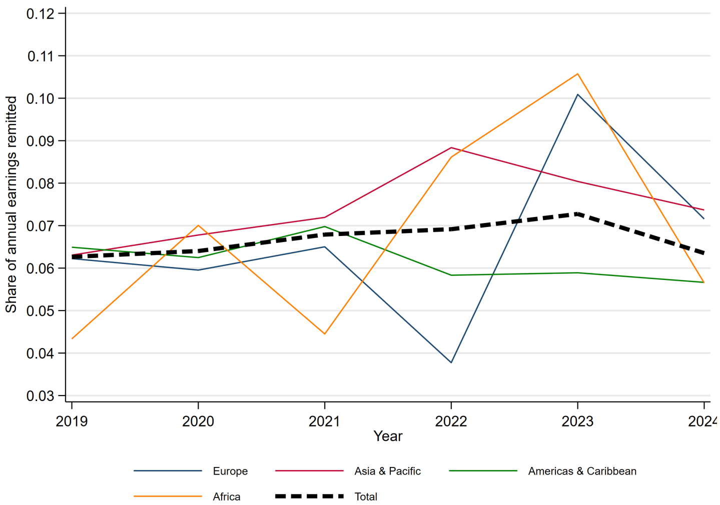
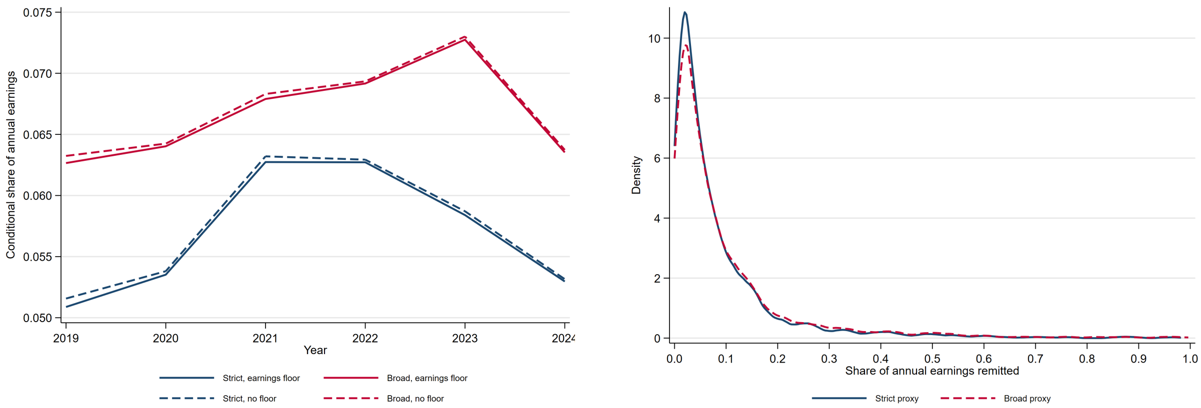


Figure 5: Evolution of Transfer Share of Earnings (Broad Proxy)

Figures 6a and 6b summarize the same design choice from aggregate and distributional perspectives. The broad proxy shifts measured shares upward and thickens the upper tail, which is informative about sensitivity but does not by itself identify a stronger structural remittance response.

Overall, interpretation remains conditional on the proxy design, and domestic-transfer contamination is treated as a first-order limitation rather than a footnote.



(a) Aggregate comparison (b) Kernel density comparison

Figure 6: Comparison of transfer shares under strict vs broad proxy definitions

4.2.1 Exchange-Rate Validation of the Proxy

The nativity-gap evidence above is necessary but not sufficient: it shows that foreign-born individuals send more, but does not confirm that the proxy captures *international* remittance behavior specifically. The key validation test exploits exchange-rate variation:

$$\Delta R_{irt} = \gamma_1 \text{Foreign-born}_i \times \Delta \log e_{rt} + \gamma_2 \Delta \log e_{rt} + X'_{irt}\Gamma + \mu_r + \lambda_t + \varepsilon_{irt}, \qquad (9)$$

where $\Delta \log e_{rt}$ is the change in a **region-level effective exchange-rate index** (constructed as a fixed-weight average of destination-country exchange rates for region r). If the proxy captures international remittance behavior, the interaction γ_1 should be significant and positive (foreign-born transfers respond to exchange-rate movements that make remitting more attractive in origin-currency terms), while US-born transfers to domestic relatives should not respond to foreign exchange-rate shocks ($\gamma_2 \approx 0$). The region-year exchange-rate panel is now integrated into the cost pipeline, so this validation regression is feasible in the next empirical pass. Under the corrected full-region migrant-stock weighting scheme, the Stage B exchange-rate coefficients are positive in the share equations but only the broad proxy is precisely estimated, while the corresponding log-amount equations turn negative.

4.3 Cost Data Construction

Transfer costs are built from the World Bank RPW micro quotes, restricted to US sender corridors and transparent quotes, collapsed to region-year means. Each region-year cell includes total cost, fee, fee share, and FX margin. The same panel now carries a region-year effective exchange-rate index built from World Development Indicators official exchange rates, aggregated with fixed 2019 migrant-stock weights from ACS 1-year Table `B05006` place-of-birth counts and geometric averaging in logs. The weighting rule is region-specific: for each region, the weights sum to 1 and each country's weight equals its share in the included 2019 US migrant stock from that region. The mapped countries for the exchange rate construction are: 30 countries for Europe, 32 for Asia-Pacific, 29 for the Americas-Caribbean and 19 for Africa. The exported weight files are sorted by migrant-stock rank within region, so the largest origin countries appear first rather than in arbitrary order. Included-country stock coverage relative to the region total is 94.7 percent for Europe, 95.5 percent for Asia-Pacific, 99.3 percent for the Americas-Caribbean and 83.2 percent for Africa.

Table 1 reports the current region-level averages used in the draft.

Table 1: RPW Region-Year Cost Summary (US Sender Corridors, 2019–2024)

Region	Mean τ_{rt}	Mean fee (USD)	Mean FX margin (%)
Europe	0.054	6.62	2.05
Asia & Pacific	0.060	8.44	1.80
Americas & Caribbean	0.058	8.11	1.74
Africa	0.060	8.04	1.97

The current data assembly supports **pre-structural validation** at region-year level. Full structural corridor elasticities would require country-level immigrant origin detail and matched corridor-year frictions which are not available.

4.4 Distance and Reduced-Form Interpretation

The remittance gravity literature does not deliver a stable distance effect: estimated coefficients can be negative, positive, or insignificant across samples and specifications (Lueth and Ruiz-Arranz, 2008; McCracken et al., 2017; De Sousa and Duval, 2010). This draft interprets distance coefficients as reduced-form friction gradients that can absorb omitted corridor composition (altruism mix, provider competition, informal channel intensity), not pure transport technology.

5 Empirical Strategy

5.1 Identification Strategy: Region-Level Exchange-Rate Variation

The identification strategy is straightforward. Transaction costs τ_{rt} change only a little within a birth region over 2019–2024, so after region and year fixed effects there is limited residual variation left in costs alone. Exchange rates, by contrast, move much more over the same period and directly change the purchasing power of a dollar remitted to origin. The paper therefore uses exchange-rate movements as the main source of within-region time variation in effective remittance cost, following the logic of Yang (2008), but aggregated to the region-year level. Define

$$\log e_{rt} \equiv \sum_{c \in r} \omega_{rc}^{0} \log e_{ct},$$

where e_{ct} is the country-c nominal exchange rate measured in local currency units per US dollar and ω_{rc}^{0} are fixed base-period weights for countries inside region r. In the current implementation, ω_{rc}^{0} are fixed 2019 migrant-stock shares by origin country within each region, drawn from ACS 1-year place-of-birth counts over the full included country list for that region rather than from the limited RPW corridor subset. Because the working SIPP files only identify broad birth regions, these weights are imported from external ACS birthplace aggregates rather than recovered from internal country-of-birth micro detail. Using fixed weights ensures that identification comes from exchange-rate movements over time rather than from changes in country composition within a region.

Define the effective dollar cost of delivering one unit of purchasing power at origin as

$$\tau_{rt}^{\text{eff}} = \frac{1 + \tau_{rt}}{e_{rt}},$$

where e_{rt} is now the region-level effective exchange-rate index. Intuitively, when origin currencies depreciate against the dollar, e_{rt} rises: each dollar sent buys more at origin, so the effective cost of delivering resources home falls.

The key identifying assumption is that, conditional on region and year fixed effects, changes in e_{rt} are unrelated to within-region changes in migrants' altruism ϕ_{ir}. This is plausible because exchange-rate movements are driven by macroeconomic shocks rather than by changes in migrants' family preferences. Empirically, $\log e_{rt}$ enters both Stage A and Stage B. In Stage B it shifts effective remittance cost, in Stage A it helps explain the participation cutoff because it changes both the purchasing power of remittances and the dollar value of origin-household resources. I do not treat it as a clean exclusion restriction in the baseline specification.

5.2 Stage A: Participation

$$\Pr(\text{send}_{irt} = 1) = \Phi(\beta_1 \log w_{irt} + \beta_2 \log(1 + \tau_{rt}) + \beta_3 \log(1 + f_{rt}) + \beta_4 \log e_{rt} + X'_{irt}\Gamma + \mu_r + \lambda_t)\,. \tag{10}$$

The exchange-rate term $\log e_{rt}$ is included throughout and provides the main within-region time-series variation for the participation decision. Because no external exclusion variable is yet available, the Stage B correction should be read as function-form assisted rather than exclusion-identified.

5.3 Stage B: Intensive Margin (Baseline)

For senders with admissible ψ_{irt}:

$$\log\left(\frac{\psi_{irt}}{1 - \psi_{irt}}\right) = \delta_1 \log \tau_{rt}^{\text{eff}} + \delta_2 \widehat{\text{IMR}}_{irt} + X'_{irt}\Pi + \mu_r + \lambda_t + \varepsilon_{irt}, \tag{11}$$

where $-\delta_1$ is the empirical counterpart to σ under the maintained structure and τ_{rt}^{eff} incorporates both the transaction cost and exchange-rate components. In the empirical implementation, the effective cost enters as

$$\log \tau_{rt}^{\text{eff}} = \log(1 + \tau_{rt}) - \log e_{rt},$$

so higher transaction costs reduce remittances while a stronger dollar against origin currencies lowers the effective cost of sending resources home. The two components can also be entered separately as a specification check, under the effective-cost interpretation, their coefficients should have equal magnitude and opposite sign.

5.4 Stage C: Aggregate Cross-Check

Region-year aggregates:

$$\log R_{rt} = a_r + b_t - \gamma \log \tau_{rt}^{\text{eff}} - \eta \log(1 + f_{rt}) + \log M_{rt} + \nu_{rt}. \tag{12}$$

Under Pareto heterogeneity calibration:

$$\sigma^G = \frac{\gamma}{\eta}, \qquad \theta^G = \gamma - \eta, \qquad \text{consistency: } \theta^G > \sigma^G - 1. \tag{13}$$

This stage is treated as a specification check, not as final structural proof, given limited support and small-cluster inference.

5.5 Stage D: Melitz Style Aggregated Moment System

In parallel to the micro equations, the paper estimates an aggregated moment system at the region-year level to reduce volatility from sparse sender cells while preserving the extensive-versus-intensive decomposition. Define:

$$N_{rt} = \sum_i 1, \qquad N^S_{rt} = \sum_i \mathbf{1}\{\text{send}_{irt} = 1\}, \qquad R_{rt} = \sum_i R_{irt}, \qquad W_{rt} = \sum_i w_{irt},$$

and moments

$$P_{rt} \equiv \frac{N^S_{rt}}{N_{rt}}, \qquad S_{rt} \equiv \frac{R_{rt}}{W_{rt}}.$$

Extensive margin (aggregate participation):

$$\text{logit}(P_{rt}) = \alpha_r + \delta_t + \pi_1 \log \tau^{\text{eff}}_{rt} + \pi_2 \log(1 + f_{rt}) + Z'_{rt}\Pi + \epsilon_{rt}, \tag{14}$$

with a count-based PPML analogue

$$\mathbb{E}[N^S_{rt} \mid X] = \exp\!\left(\alpha_r + \delta_t + \pi_1 \log \tau^{\text{eff}}_{rt} + \pi_2 \log(1 + f_{rt}) + \log N_{rt}\right).$$

Intensive margin (aggregate conditional transfer pressure):

$$\log S_{rt} = c_r + d_t + \rho_1 \log \tau^{\text{eff}}_{rt} + \rho_2 \log(1 + f_{rt}) + Z'_{rt}\varrho + u_{rt}, \tag{15}$$

with the corresponding level specification

$$\mathbb{E}[R_{rt} \mid X] = \exp\!\left(c_r + d_t + \rho_1 \log \tau^{\text{eff}}_{rt} + \rho_2 \log(1 + f_{rt}) + \log W_{rt}\right).$$

This aggregation layer is not a replacement for micro identification. Its role is to impose the same Melitz/HMR-style margin decomposition on stable moments, diagnose whether micro coefficients are reflected in aggregate behavior, and reduce visual and inferential instability from thin region-year sender cells. The cost is reduced control for within-region composition, so all aggregated estimates are interpreted as disciplined robustness checks rather than primary structural evidence.

6 Current Empirical Status

6.1 Descriptive Facts

Using the harmonized bounded sample and ratio-of-totals metric, conditional transfer shares are materially below the 15% benchmark and cluster around 5–6% for foreign-born senders in 2019–2024, with US-born shares lower but non-zero. This is consistent with a proxy that captures remittance-relevant behavior plus domestic-transfer noise.

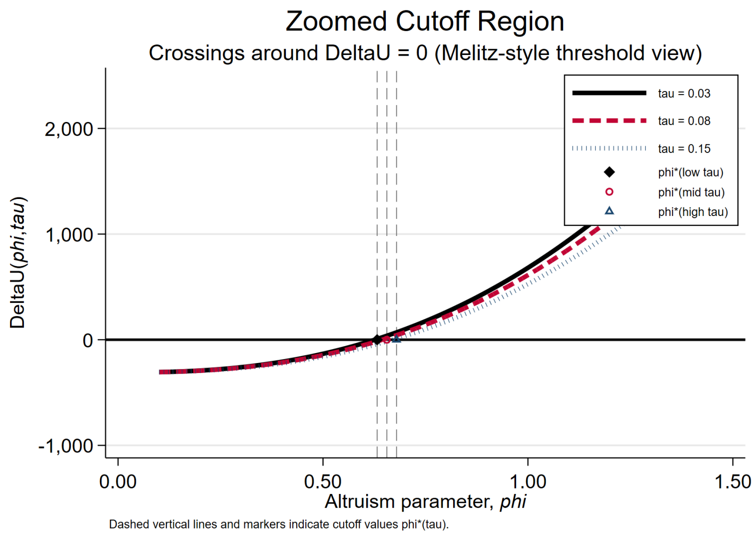


Figure 7: Stage A Intuition (Simulated): Participation Cutoff under Higher Remittance Frictions

Figure 7 is a model-implied simulation (not an estimated reduced-form plot). It documents the comparative statics used to interpret Stage A, namely that higher τ_{rt} or f_{rt} shifts the participation cutoff upward under the maintained CES structure.

6.2 Structural Diagnostics

Current estimates are not yet structurally stable. Once the diagnostics are defined on effective cost rather than on transaction fees alone, the quantitative implications change materially. Exchange-rate integration matters because $\log e_{rt}$ is observed for 98.7 percent of the mapped foreign-born sample and generates substantial within-region time variation in $\log \tau_{rt}^{\text{eff}}$. In the current run, the maximum within-region standard deviation of $\log \tau_{rt}^{\text{eff}}$ is 0.2597, well above the identification threshold used in the estimation code. As a consequence:

- Stage B raw implied elasticities are estimated at $\hat{\sigma}_{\text{raw}} = 0.976$ for the strict proxy and $\hat{\sigma}_{\text{raw}} = 1.206$ for the broad proxy.
- The year-specific σ_t decomposition lies in a much narrower range: strict-proxy estimates range from 1.18 to 1.51 across years, while broad-proxy estimates range from 1.00 to 1.25. This contrasts with the much wider dispersion in the only τ diagnostic.
- The log-amount robustness is still weak. The effective-cost coefficient is small and wrong-signed in both proxies (0.013 strict, 0.080 broad), so the cleanest structural evidence still comes from the logit-share equation rather than the amount regression.

- Stage C gravity elasticities (γ, η) are numerically identified, but they do not yet provide a reassuring structural cross-check. In the strict proxy, $\hat{\theta}^G < \hat{\sigma}^G - 1$ fails the Pareto consistency condition. In the broad proxy, the fixed-cost coefficient has the wrong sign ($\hat{\eta} < 0$) even though the mechanical Pareto check passes.
- No exclusion variable is yet available for Stage A, so the IMR correction is still identified by functional form rather than by an external participation shifter.
- Stage A and Stage B rely on 4–6 effective clusters (birth regions), far below the cluster counts typically recommended by Cameron and Miller (2015) for reliable cluster-robust inference. All p-values are diagnostic only.

These diagnostics indicate that exchange-rate increases affect the identifying variation materially: once the structural object is defined as $\log \tau_{rt}^{\text{eff}}$, the intensive-margin elasticity is no longer tied mechanically to near-zero variation in transaction fees alone. The migrant-stock reweighting is useful precisely because it disciplines the economic object being measured. What is still missing is a stronger exclusion strategy, richer origin disaggregation, and more effective clusters. The current results are best read as structural evidence on the intensive margin, but not yet as a complete structural design.

6.3 Preliminary Elasticity and Altruism Estimates

Table 2 reports the current Stage B outputs once the specification is written in terms of effective cost. Both proxies now deliver raw elasticities close to one, and neither requires calibration for reporting.

Table 2: Current Stage B Effective-Cost Outputs

Proxy	$\hat{\sigma}_{\text{raw}}$	$\hat{\sigma}_{\text{reported}}$	Calibrated	$\hat{\beta}^{amount}_{\tau^{\text{eff}}}$	N_{share}	N_{amount}	$\max_r \text{sd}[\log \tau_{rt}^{\text{eff}}]$
Strict	0.976	0.976	No	0.013	2204	2207	0.2597
Broad	1.206	1.206	No	0.080	2517	2524	0.2597

The year-by-year σ_t decomposition is now informative rather than pathological, but Stage C gravity elasticities still do not line up cleanly with the micro evidence.

Using the estimated Stage B elasticities, the corresponding relative altruism estimates (normalized to Europe = 1) are:

Table 3: Relative Altruism by Region (Current Run, Europe = 1)

Region	Strict proxy $\hat{\phi}_r/\hat{\phi}_{Europe}$	Broad proxy $\hat{\phi}_r/\hat{\phi}_{Europe}$
Europe	1.000	1.000
Asia & Pacific	1.004	1.498
Americas & Caribbean	1.560	1.840
Africa	0.584	0.855

Interpretation. Once the diagnostics are computed on $\log \tau_{rt}^{\text{eff}}$ rather than on $\log(1+\tau_{rt})$ alone, the intensive-margin estimates fall in a narrower and more interpretable range. The implied elasticity of substitution is close to one in both proxies, which suggests that local consumption and origin-household resources are neither near-perfect substitutes nor near-Leontief complements. This change is substantive rather than cosmetic: the estimated σ is no longer being forced by calibration, and the year-by-year decomposition no longer exhibits extreme values. At the same time, the broader structural design is still incomplete. The Stage A correction remains function-form identified, cluster counts remain very small, and Stage C still fails to validate the micro estimates cleanly. The altruism rankings are therefore informative but still conditional on a fragile empirical environment, especially for Africa where the strict and broad proxies remain farther apart than is ideal. The right reading is that the project now has a plausible intensive-margin structural estimate, but it still needs a stronger participation shifter and a more convincing aggregate cross-check before strong policy counterfactuals are warranted.

7 Counterfactual Agenda and Scope

Planned policy exercises:

1. Reduce average remittance costs toward SDG 10.c benchmark.
2. Decompose effects into extensive vs intensive margins.
3. Evaluate fixed-cost digitalization scenarios.
4. Quantify heterogeneity by migrant type and origin region.
5. Labour market shock scenario

Scope condition for credible counterfactuals: only after the effective-cost specification is paired with a stronger exclusion strategy, a stable aggregate cross-check, and ideally finer origin/corridor variation. Until then, counterfactuals are best presented as sensitivity exercises around the estimated σ range in the current run, not as final policy elasticities.

8 Roadmap

1. **Priority:** Implement the exchange-rate proxy validation (foreign-born vs. US-born differential response to exchange-rate shocks) to establish that the transfer proxy captures international remittance behavior.
2. **Priority:** Add a genuine exclusion variable for Stage A. At present, the IMR correction is function-form identified only, which is too weak for strong structural claims.
3. **Priority:** Diagnose why the amount and aggregate cross-checks still fail to line up with the share equation under the effective-cost specification, in practice this

means testing alternative dependent-variable parameterizations, checking influential region-year cells, and understanding the wrong-signed fixed-cost coefficient in the broad Stage C regression.

4. **Near-term:** Estimate recipient-type-specific participation equations (Section 3.3) to test the multi-destination pecking-order prediction.
5. **Preferred extension (data permitting):** Build country origin mapping from micro birthplace detail or linked external micro source to increase the number of cost clusters from 4 to 30+.
6. Retain strict proxy as baseline, present broad proxy as sensitivity envelope.

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